



DSP Integrated Circuits

Chapter 2: Review of DSP Basic Theory Part II

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Contents

- ◇ Sampling theorem
- ◇ Z-transform
- ◇ Discrete-time system



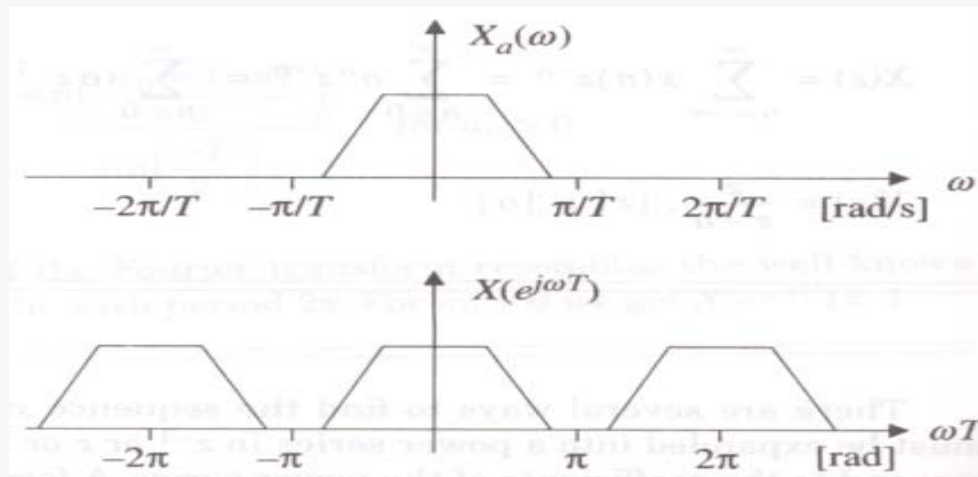


Sampling theorem

Sampling of Analog Signals (1)

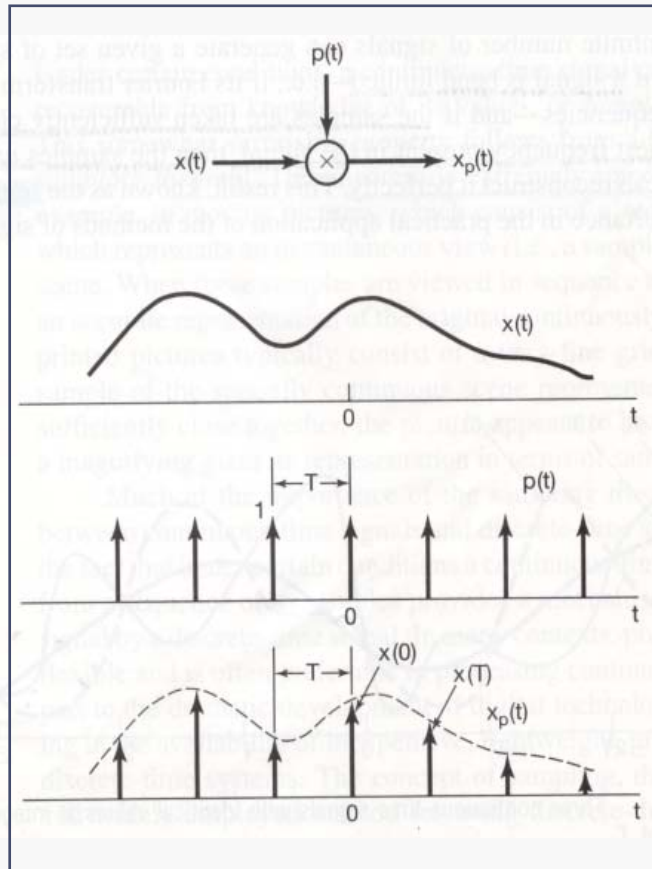
- Fourier transform of sequence $x(n)$ that is obtained by sampling the analog signal $x_a(t)$ at $t=nT$, is related to the Fourier transform $X_a(\omega)$ of analog signal $x_a(t)$ by Poisson's summation formula :

$$X(e^{j\omega T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a(\omega + \frac{2\pi k}{T})$$

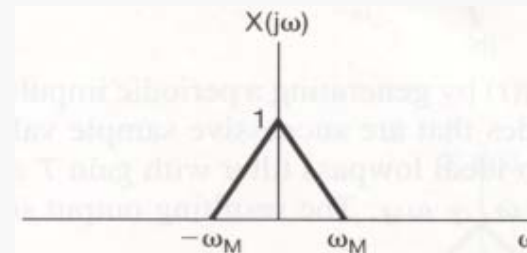


- If analog signal is not bandlimited or if sampling frequency is too low, the spectrum images will be overlapped (cannot be recovered)

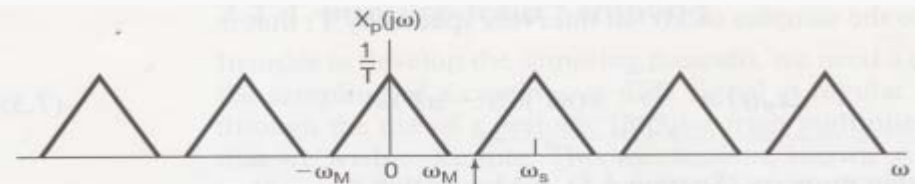
Sampling of Analog Signals (2)



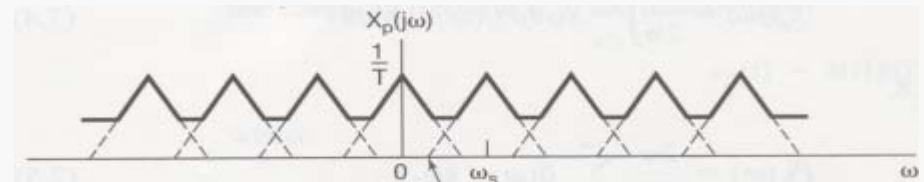
<Impulse-train Sampling in TD>



(a) Spectrum of analog signal



(c) Spectrum of sampled signal for $W_s > 2W_M$



(d) Spectrum of sampled signal for $W_s < 2W_M$

<Spectrum of analog & Sampled signals>

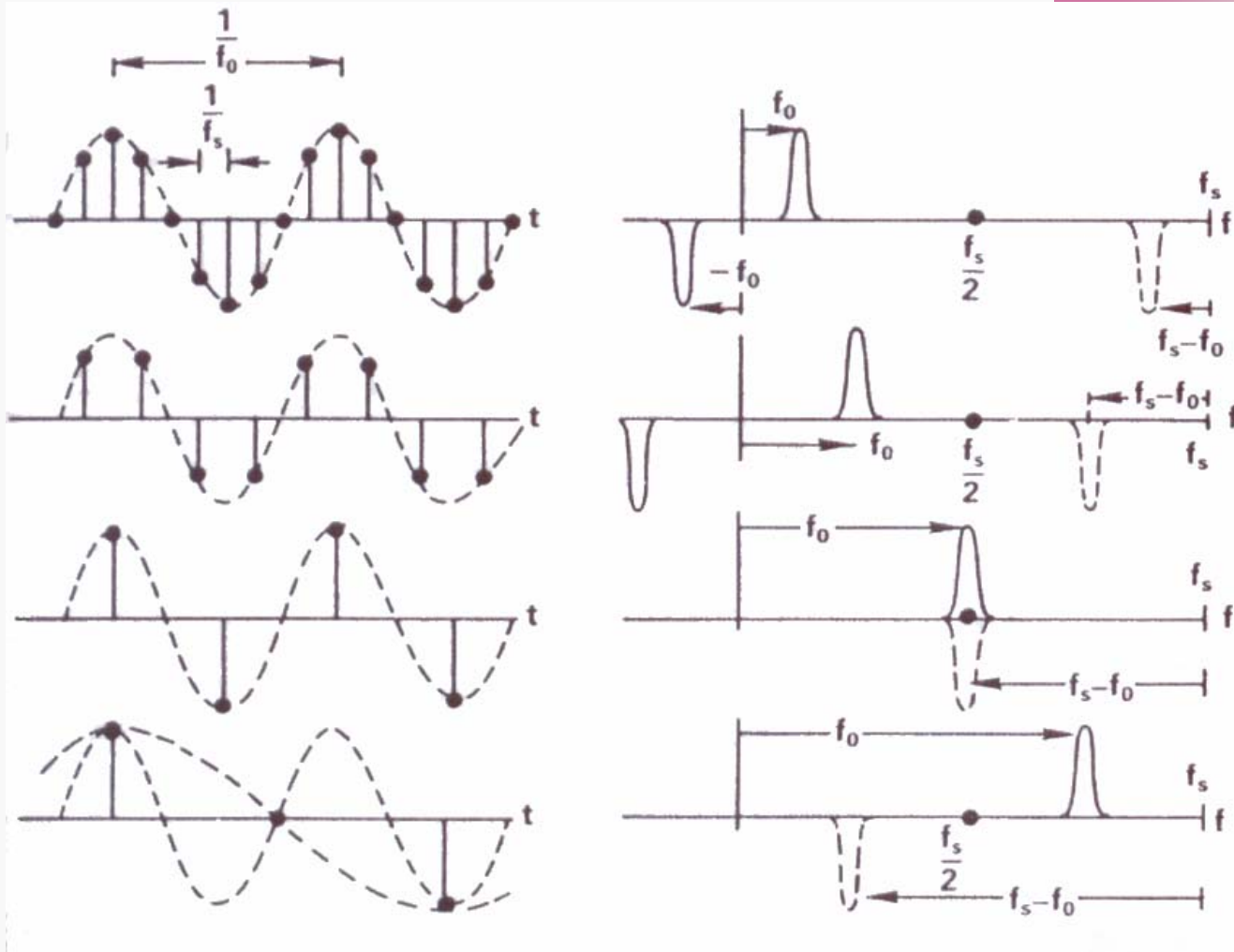
Nyquist Sampling Theorem

- ◇ If an analog signal, $x_a(t)$, is bandlimited so that $|X_a(\omega)| = 0$ for $|\omega| \geq \omega_0$, all information is retained in the sequence $x(nT)$, obtained by periodic sampling $x_a(t)$ at $t = nT$ where $T < \pi / \omega_0$
(or if sampling frequency $\omega_s > 2\omega_0$ where $\omega_s = 2\pi / T$)
 - The sampling rate must be larger than twice the signal bandwidth to recover the original information exactly from the sampled sequence
 - $2\omega_0$: Nyquist sampling rate

- ◇ Aliasing : Effect of undersampling
 - when $\omega_s < 2\omega_0$, the spectrum of $x_a(t)$ is no longer replicated in $X(e^{j\omega T})$, which is the spectrum of $x(n)$, and thus no longer recoverable by lowpass filtering



Sampling & aliasing



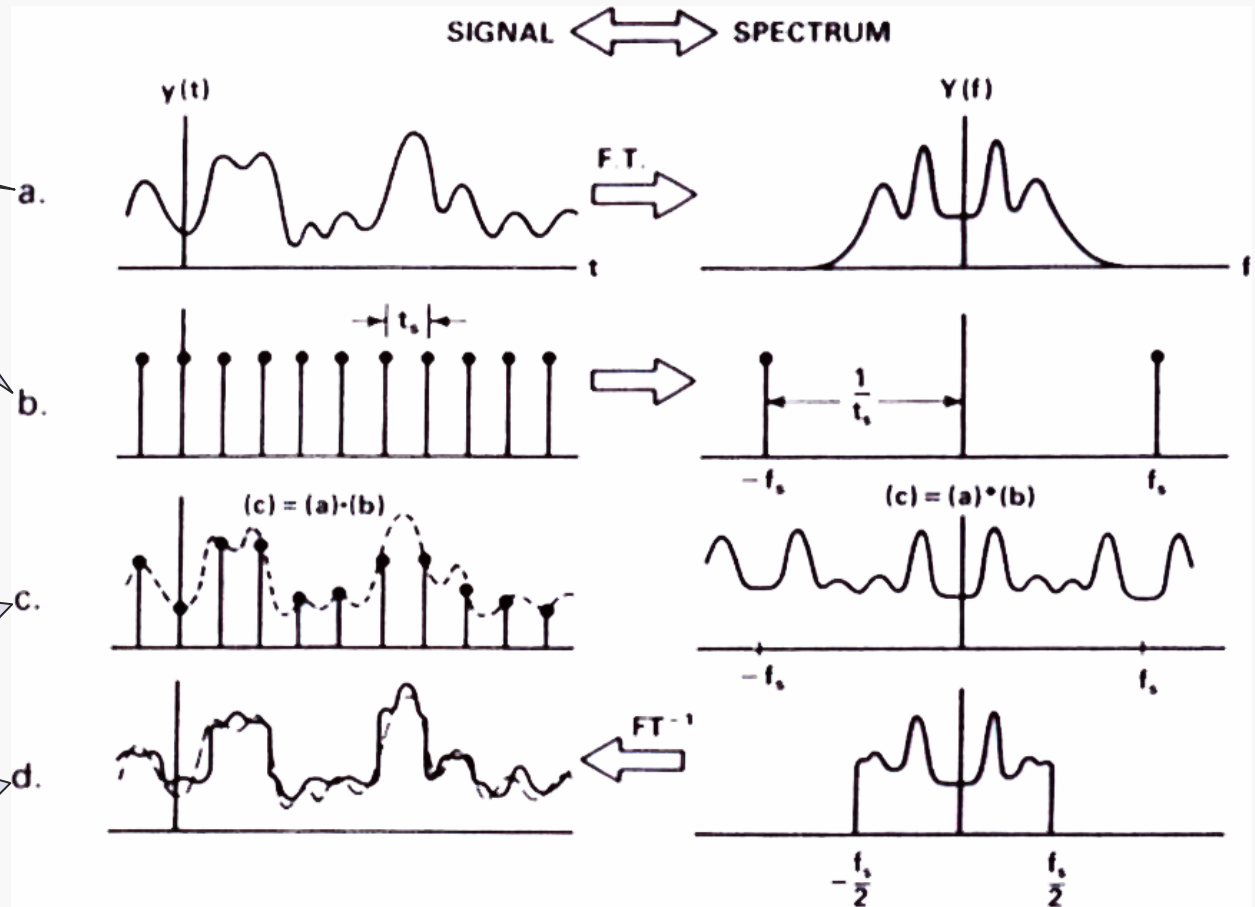
Spectral features of sampled data(1)

Given continuous function $y(t)$

Sampling impulse at intervals t_s

Fourier transform (F.T) of the sampled data = Convolution of F.T (a) with F.T (b)

Inverse transform of $Y(f)$ which have some errors due to the overlap in (c)'s freq. spectrum

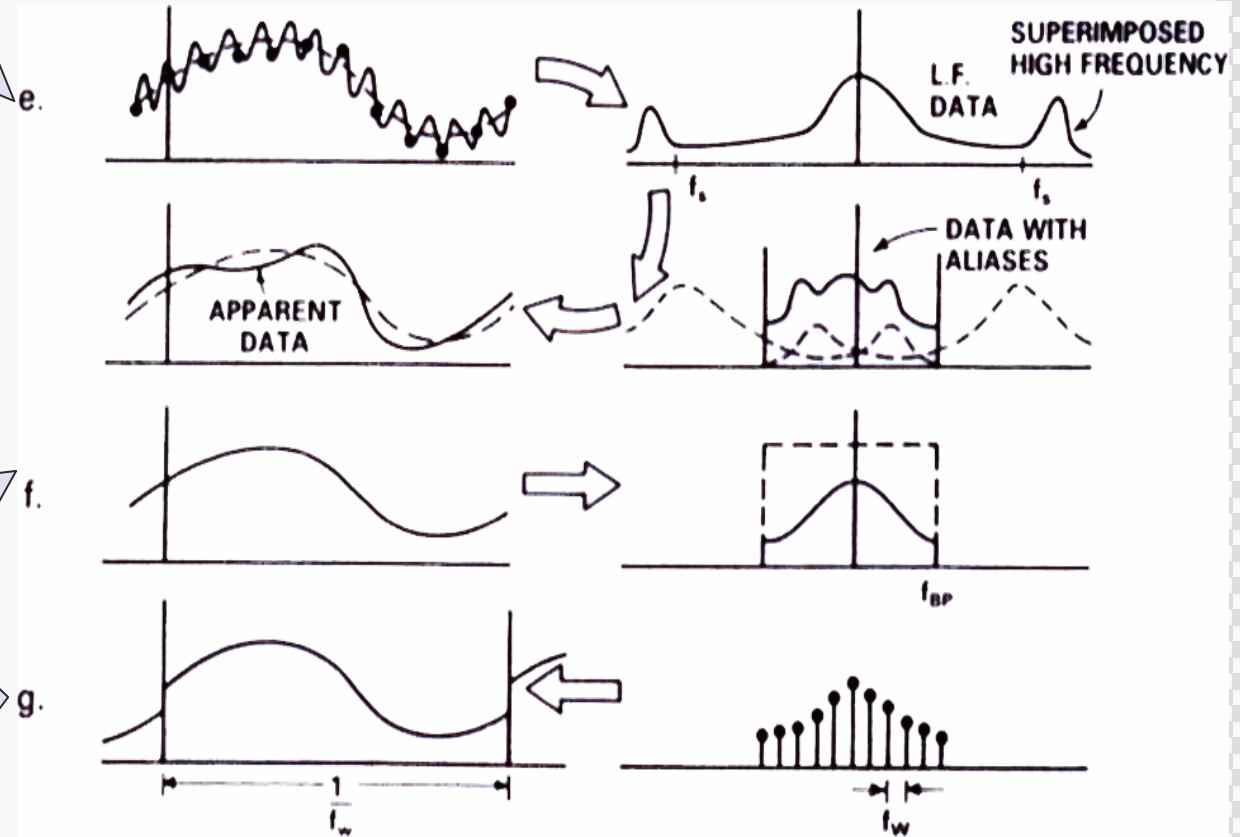


Spectral features of sampled data(2)

- $y(t)$ has some noise component at a freq. above $f_s/2$
- The alias is very real and troublesome to track down

- Lowpass filtering prior to doing transform to limit the spectral content
- No alias terms occur

If the freq. spectrum is sampled and an insufficient # of samples are used, discontinuities will be introduced at the boundary





Z-transform

Laplace Transform(1)

◇ Forward Transform

$$F(S) = \int_0^{+\infty} e^{-st} f(t) ds$$

◇ Inverse Transform

$$f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} e^{st} F(s) ds$$

where $s = \sigma + j\omega$



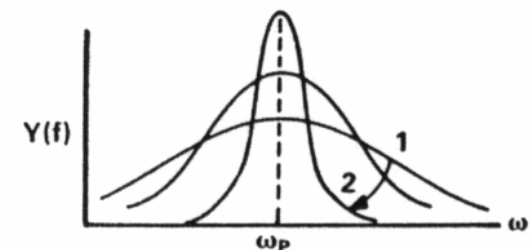
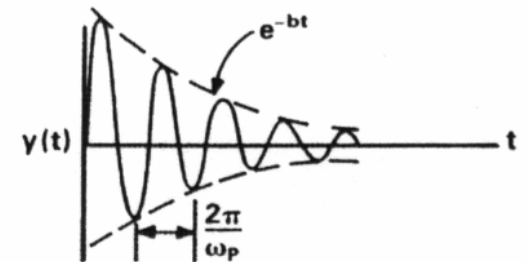
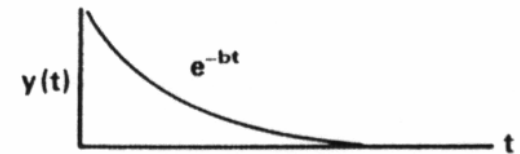
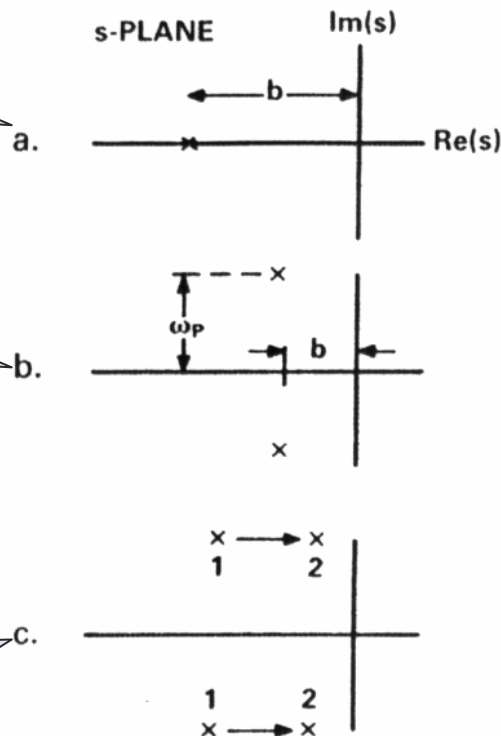
Laplace Transform(2)

◆ Poles and corresponding impulse response for simple analog system

- First order
= RC low pass

- Second order
= Resonant system
with damping

- Sharpness of spectral
response
→ A function of the
closeness of poles to the
vertical axis, $\text{Im}(s)=0$



Z-Transform (1)

- ◇ Z-transform : discrete-time counterpart of Laplace transform which is a generalization of continuous-time Fourier transform

■ useful to analysis of stability of system

- ◇ Z-transform :

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}, R_+ < |z| < R_-$$

where z : Complex variable $(= r(e^{j\omega}))$

$R_+ < |z| < R_-$: Region of convergence

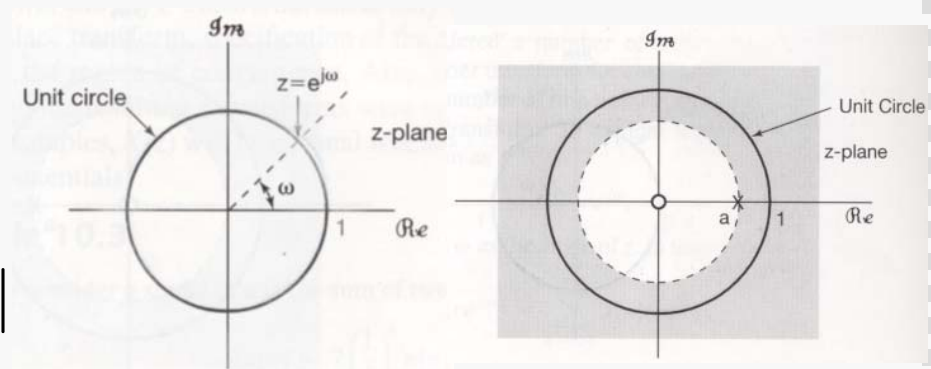
- ◇ Z-transform reduces to Fourier transform when $z = e^{j\omega T}$
(on the unit circle in z-plane)

■ Analogy : Laplace transform reduces to Fourier transform on imaginary axis (i.e. for $s=j\omega$)

■ For the step sequence

$$x(n) = \begin{cases} a_n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, |z| > |a|$$



Z-Transform (2)

- ◇ Time Shifting property of a sequence :

$$x(n - k) \leftrightarrow X(z)z^{-k}$$

- ◇ Convolution property

$$x_1[n] * x_2[n] \leftrightarrow X_1(z) \cdot X_2(z)$$



Z-Transform (3)

◇ Inverse Z-transform

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

where C : contour inside the region of convergence



Region of Convergence (ROC)

$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} \right| = \left| \sum_{n=-\infty}^{\infty} x[n] \cdot (re^{j\omega})^n \right| \leq \left| \sum_{n=-\infty}^{\infty} x[n] \right| r^n$$

- ◆ The ROC does not contain any poles
- ◆ The ROC is the connected region
- ◆ If $x[n]$ is finite duration, the ROC is the entire z -plane, except $z=0$
- ◆ If $x[n]$ is right sided, the ROC is the outside of the outermost poles
- ◆ If $x[n]$ is left sided, the ROC is the inside of the innermost poles
- ◆ If $x[n]$ is both sided, the ROC must have an inter-section
- ◆ If unit circle $\subset$ ROC, the system is stable
- ◆ If unit circle $\not\subset$ ROC, the system is unstable



ROC (2)

◇ Example

- Right-sided sequence; casual

$$x[n] = a^n \cdot u[n]$$

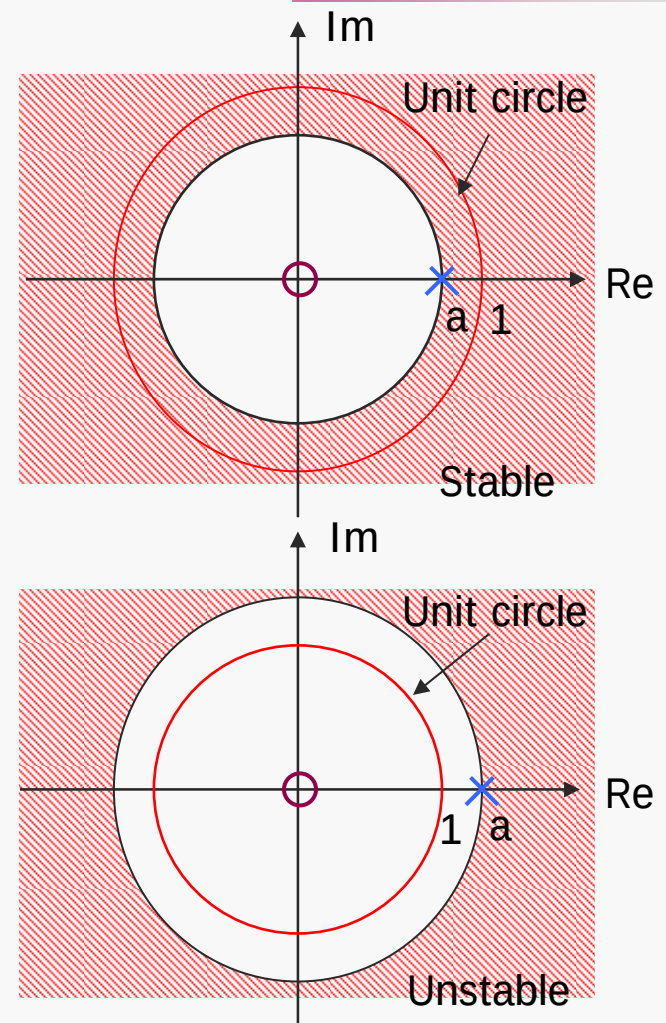
$$X(z) = \sum_{n=0}^{\infty} a^n \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{z} \right)^n$$

$$= \frac{1}{1 - a/z}$$

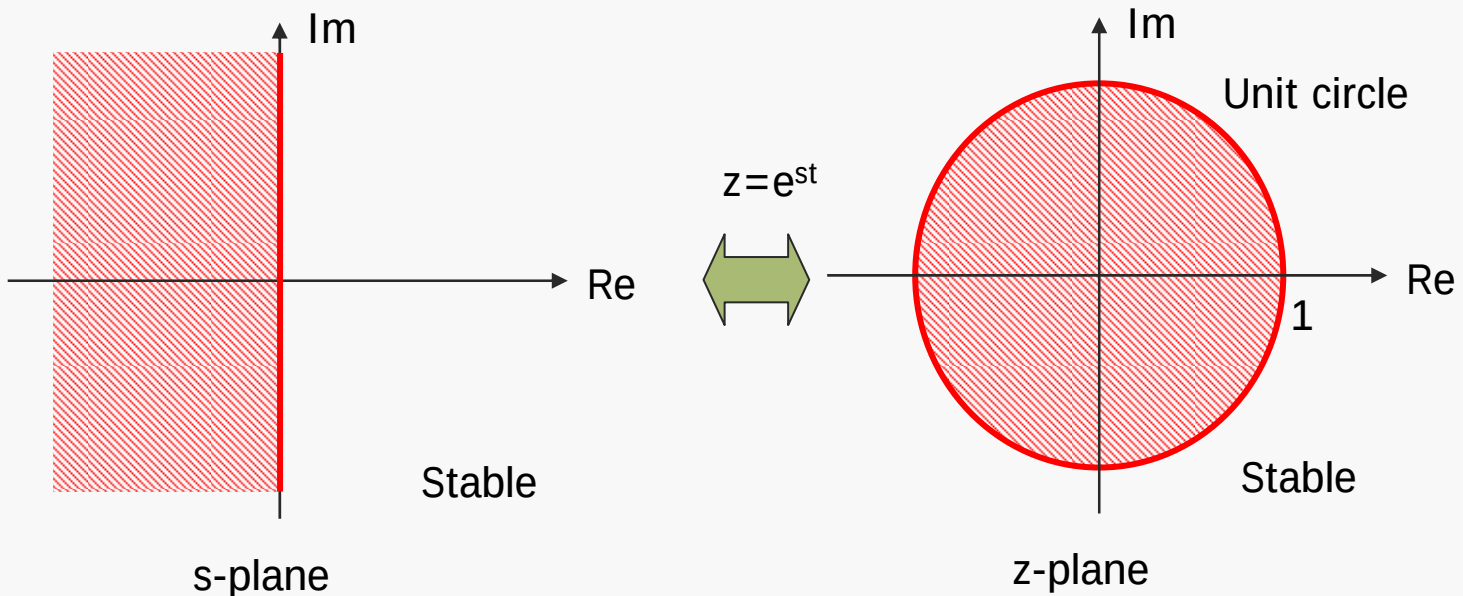
$$= \frac{z}{z - a}$$

$$\text{ROC} ; |z| > a$$



Z-transform & Laplace Transform

- ◇ The laplace transform, along the frequency axis, $s=j\omega$, is the z-transform evaluated on the unit circle, $z=1$
- The left half of s-plane maps within the unit circle of the z-plane
- A distance, $1/t_s$, along the real frequency axis wraps once around the unit circle in z



Properties of Z-transform(1)

◇ Linearity

$$ax_1[n] + bx_2[n] \Rightarrow aX_1(z) + bX_2(z) \quad \text{ROC}; R_1 \cap R_2$$

◇ Time shifting property

$$x[n - n_0] \Rightarrow \sum_{n=-\infty}^{\infty} x[n - n_0] z^{-n} = \sum_{n=-\infty}^{\infty} x[n] z^{-n - n_0} = z^{-n_0} \cdot X(z)$$

◇ Multiplication by exponential sequences

$$z_0^n x[n] \Rightarrow \sum_{n=-\infty}^{\infty} x[n] z_0^n z^{-n} = \sum_{n=-\infty}^{\infty} x[n] \left(\frac{z_0}{z} \right)^n = X\left(\frac{z}{z_0} \right)$$

In this case, the ROC is changed. $|z| < a \rightarrow \left| \frac{z}{z_0} \right| < a, |z| < a|z_0|$



Properties of Z-transform(2)

- ◆ Differentiation of $X(z)$

$$nx[n] \Rightarrow -z \frac{dX(z)}{dz}$$

- ◆ Conjugation

$$x^*[n] \Rightarrow X^*(z^*)$$

- ◆ Time reversal

$$x[-n] \Rightarrow X\left(\frac{1}{z}\right) \quad \text{ROC; } \frac{1}{R}$$

- ◆ Convolution

$$x_1[n] * x_2[n] \Rightarrow X_1(z)X_2(z) \quad \text{ROC; } R_1 \cap R_2$$





Discrete-time System

Difference Equations (1)

- ◇ Many discrete-time digital systems (such as digital filters) can be described by difference equations with constant coefficients

- ◇ I/O relation for Nth-order LSI System :

$$y(n) = \sum_{k=1}^N b_k y(n-k) + \sum_{k=0}^M a_k x(n-k)$$

- ◇ Two different types of impulse response

- 1) FIR (Finite-length Impulse Response) system :

- Impulse response becomes zero after a finite number of samples
- If $b_k=0$ for all k
- Realized by using non-recursive type because of stability problem

- 2) IIR (Infinite-length Impulse Response) system :

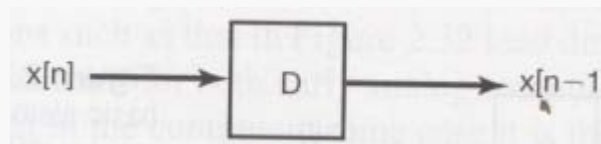
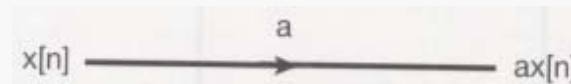
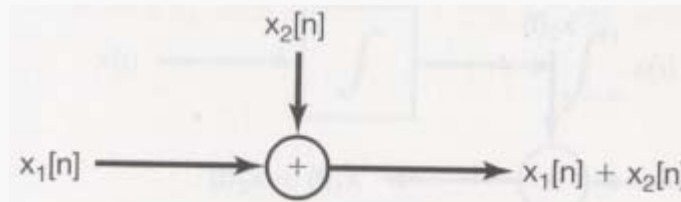
- Impulse response decays toward zero, but never reaches zero
- If $b_k \neq 0$ for at least one k
- Realized by using recursive algorithms



Difference Equations (2)

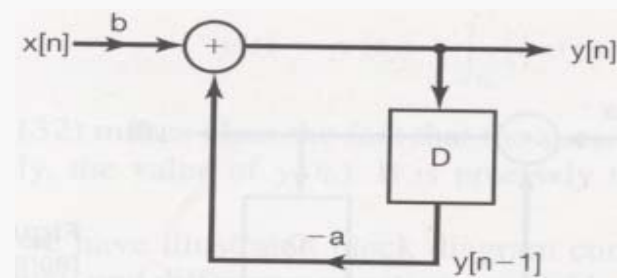
◇ Block diagram Representation of difference equation

■ Basic elements



■ Example for 1st-order difference equation

$$y[n] = -ay[n-1] + bx[n]$$



Frequency Response

- ◇ Frequency response of LTI system, $H(e^{j\omega T})$, obtained with complex sinusoidal input signal, $x(n) = e^{jn\omega T}$, is

$$\begin{aligned} y(e^{jn\omega T}) &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \sum_{k=-\infty}^{\infty} h(k)e^{j\omega(n-k)T} + \sum_{k=-\infty}^{\infty} h(k)e^{-jk\omega T}e^{jn\omega T} \\ &= H(e^{j\omega T})e^{jn\omega T} = H(e^{j\omega T})x(n) \end{aligned}$$

where $H(e^{j\omega T}) = \sum_{k=-\infty}^{\infty} h(k) \cdot e^{-jk\omega T} = |H(e^{j\omega T})| \cdot e^{j\Phi(\omega T)}$

$|H(e^{j\omega T})|$: magnitude response

$A(\omega T) \triangleq -20\log_{10}(|H(e^{j\omega T})|)$ (dB) : Attenuation function

$\Phi(\omega T)$: phase response

$$\triangleq \arg\{H(e^{j\omega T})\} = \tan^{-1} \left\{ \frac{\text{Im}\{H(e^{j\omega T})\}}{\text{Re}\{H(e^{j\omega T})\}} \right\}$$



Transfer function

- Transfer function, $H(z)$

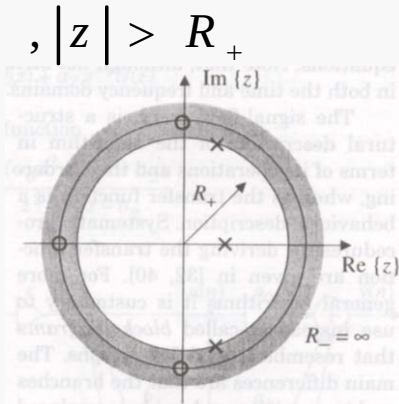
$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M a_k z^{-k}}{1 - \sum_{k=1}^N b_k z^{-k}}$$

- Example of transfer function for digital lowpass filter

$$H(z) = \frac{0.18675628(z+1)(z^2 + 0.09021784z + 1)}{(z - 0.36278303)(z^2 - 0.51108872z + 0.73629559)}$$

$$R_+ = \sqrt{0.73629559} = 0.85807668$$

- Zeros : roots of numerator
- Poles : roots of denominator



- For a causal stable system, poles are constrained to be inside the unit circle while zeros can be located anywhere

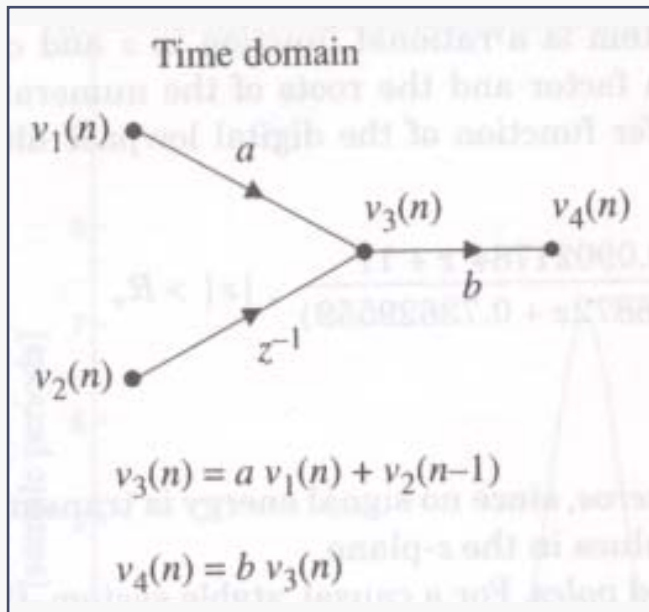
Signal-Flow Graphs (1)

- ◇ Digital filter algorithm : described by a system of difference equations
- ◇ Signal-Flow Graphs (SFG) :
 - Alternative representation which more clearly illustrates computational properties of an algorithm than difference eq.
 - A structural description of algorithm in terms of its operations and their ordering
- ◇ Basic elements :
 - Nodes : represent variables (signals)
 - Directed branches : represent relations between the variables (nodes) with associate transmittances
 - Value of a node : sum of signal values (node values multiplied by corresponding transmittances) entering the node

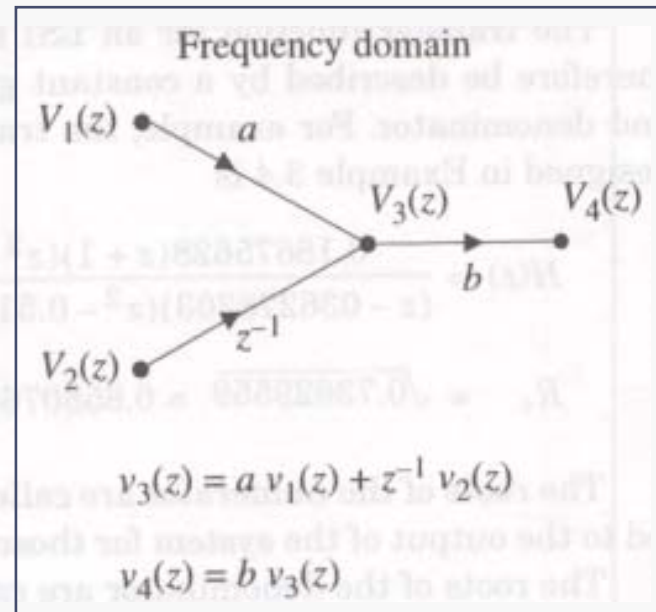


Signal-flow Graphs (2)

<Direct form I >



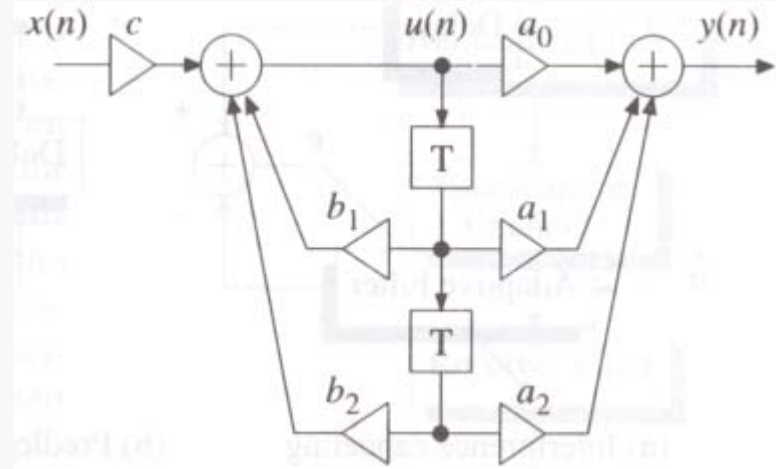
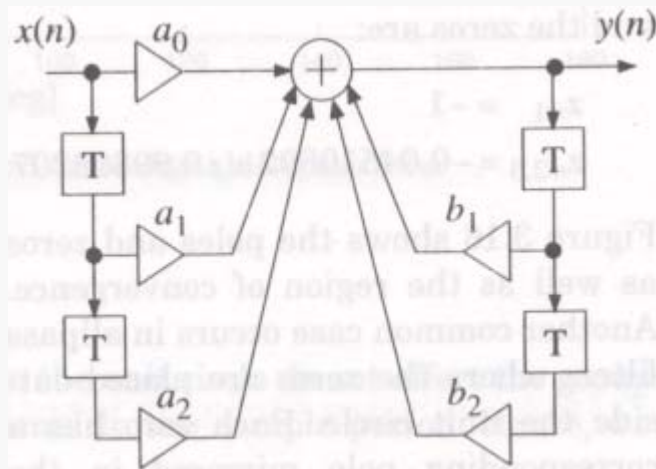
<Direct form II >



- ◇ For more general algorithms, it is common to use block diagram which contains discrete blocks and explicit summation node

Filter Structures

- Typical block diagrams of digital filter structures



Where T (or z^{-1}) : delay element

- Most important criterion for selecting a filter structure :
 - Robustness (always should be stable even under nonlinear conditions)